Development of Group Re-Identification System

Analyzing Human Groups in Public Spaces

By

Serikbay Arsen
Aidos Zhasulan

School of Information Technologies
Astana IT University

6B06107 — Mathematical and
Computational Science
Supervisor: Miras Mussabek

June 2026
Astana

Abstract

This diploma work investigates the development of a group re-identification system for analyzing human groups in public spaces. The study focuses on the problem of matching the same social group across non-overlapping camera views under changes in viewpoint, occlusion, group topology, and member composition. The work combines a review of recent group re-identification methods with the design of a system-oriented pipeline that supports detection, group representation, cross-camera matching, and quantitative evaluation.

Keywords: group re-identification, person re-identification, computer vision, video analytics, public-space analysis

Dedication and acknowledgements

I would like to dedicate this work to ...

Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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Definitions

The following terms are used in this diploma work:

Re-identification	The task of matching the same target observed by different cameras or at different times.
Person re-identification	Re-identification of an individual person based on appearance features and cross-camera matching.
Group re-identification	Re-identification of a human group viewed across non-overlapping cameras while accounting for member interactions, spatial layout, and composition changes.
Topology	The relative arrangement and structural organization of group members in the scene.
Occlusion	Partial or full visual blockage of a target by other people, objects, or scene geometry.
Embedding	A numerical feature representation used to compare images, persons, or groups in a common feature space.

Designations and Abbreviations

The following designations and abbreviations are used in this diploma work:

AI	Artificial Intelligence
CNN	Convolutional Neural Network
G-ReID	Group Re-Identification
GNN	Graph Neural Network
ML	Machine Learning
mAP	Mean Average Precision
ReID	Re-Identification
Rank-1	Rank-1 retrieval accuracy

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Chapter 1

Introduction

1.1 Relevance of the Topic

Video surveillance and public-space monitoring have become important elements of urban security, crowd analysis, and smart-city infrastructure. In many practical scenarios, analysts are interested not only in isolated individuals but also in the movement and behavior of human groups. Group-level analysis is relevant for understanding social interactions, monitoring coordinated activities, and improving event reconstruction across distributed camera networks. In security-oriented applications, the ability to follow a group across non-overlapping cameras may support the analysis of coordinated behavior, anomaly detection, and post-event investigation. In smart-city environments, group-aware visual analytics can also contribute to pedestrian flow monitoring, crowd management, and the study of social dynamics in public spaces [2].

At the same time, many existing surveillance pipelines are centered on the individual and rely on person re-identification as the main retrieval mechanism. Although person re-identification has achieved substantial progress, it does not fully address scenarios in which the object of interest is a social unit rather than a single pedestrian. A group cannot be reduced to an independent collection of isolated identities, because its semantics are shaped not only by the appearance of its members but also by the relationships among them, their relative arrangement, and their temporal stability across views [2, 3].

1.2 Problem Statement

Standard person re-identification methods are designed to match a single pedestrian across cameras. However, these methods are insufficient when the object of interest is a social group whose members change positions, partially occlude one another, or vary across observations. Therefore, the problem considered in this work is the development of a system for group re-identification in public spaces under realistic cross-camera conditions.

Group re-identification is substantially more challenging than conventional person re-identification because the target is not a single, visually bounded individual, but a dynamic social unit observed across different cameras. In standard person re-identification, the task is to match one pedestrian despite variations in pose, illumination, viewpoint, and partial occlusion. In contrast, group re-identification must simultaneously capture the appearance of multiple individuals and the relationships among them, while remaining robust to changes in group composition and spatial arrangement. As a result, the problem extends beyond individual visual matching and requires modeling the collective structure of the group as a flexible and evolving entity [2, 3].

One major source of difficulty is the variation in group topology and spatial layout. A human group does not preserve a rigid geometric structure while moving through public space. The relative positions of members may change as people walk, interact, avoid obstacles, or respond to surrounding conditions. The same group may appear as a compact cluster in one camera and as a loose line formation in another. These changes are further amplified by viewpoint differences across non-overlapping cameras, making two-dimensional spatial arrangements unreliable indicators of group identity. Consequently, a group cannot be represented as a fixed configuration, and robust re-identification requires methods capable of handling non-rigid and ambiguous spatial structures [4, 5].

Another challenge is member variation, or the compositional fluidity of groups. Unlike an individual person, a group may not remain identical over time: some members may temporarily leave, others may join, and some may be only partially visible in one camera view. Therefore, the identity of a group is often preserved even when not all members are consistently present. This makes group re-identification inherently tolerant to partial mismatch, but also more difficult to formalize and learn. A successful model must distinguish meaningful changes in group composition from irrelevant noise caused by nearby pedestrians, fragmented detections, or temporary interactions with unrelated

individuals. In addition, not all members contribute equally to group identity; some individuals may form a stable core of the group, while others are more peripheral and less informative [1, 5].

Intra-group occlusion further increases the complexity of the task. Because group members move in close proximity, they frequently obscure one another, especially in crowded public scenes. Partial or full occlusion can hide important appearance cues such as clothing patterns, body shape, or carried objects, reducing the reliability of individual-level descriptors. At the same time, group images often contain larger spatial regions than single-person crops, which makes them more vulnerable to background clutter and irrelevant contextual information. As a result, methods that rely only on global visual features may fail to isolate the truly discriminative characteristics of the group [2, 3].

Finally, group re-identification requires the integration of local and global information. A model must preserve fine-grained person-level appearance cues while also reasoning about the overall group structure and social context. This dual requirement makes the task fundamentally more complex than person re-identification. In person ReID, the object of interest is a single identity; in group ReID, the system must determine whether two sets of people, observed under changing spatial, visual, and compositional conditions, correspond to the same social unit. For this reason, topology changes, occlusion, and member variation make group re-identification a substantially more difficult and less constrained problem than traditional person re-identification [2, 3].

In formal terms, group re-identification can be defined as the task of associating and retrieving group images containing the same members across disjoint, non-overlapping camera views. Let a probe set be denoted by $\mathcal{Q} = \{I_q\}$ and a gallery set by $\mathcal{G} = \{I_g\}$, where each image corresponds to an observation of a human group captured by a particular camera. For a given probe image I_q, the objective is to rank all gallery images according to their similarity and to retrieve those that correspond to the same group identity.

Each group image I is represented as a set of detected persons $S_I = \{p_1, p_2, \ldots, p_N\}$, where N is the number of visible members in the image. Each member p_i is associated with a bounding box $b_i = (x_i, y_i, w_i, h_i)$ that specifies the position and size of the person in the image plane. If additional annotation is available, the member may also have an identity label y_i. Therefore, the input to the group re-identification system may include: (1) a group image, (2) a set of bounding boxes for its detected members, and (3)

member identities when such labels are available in the dataset or annotation protocol. On the basis of these inputs, the system must learn a representation that combines local appearance descriptors with group-level structural information.

The output of the system is a group-level matching result across cameras. In practice, this is typically expressed as a ranked list of gallery candidates for each probe image or as a predicted group identity label. Two observations from different cameras are considered a positive match if they correspond to the same group according to the dataset definition. In many group re-identification benchmarks, this condition is formulated through a membership overlap rule, where two group images are treated as the same class if a sufficient proportion of members is shared between them [4, 6].

Depending on the annotation setting, the problem may be studied under fully-supervised or weakly-supervised conditions. In the fully-supervised setting, both group-level labels and person-level member identities are available. In the weakly-supervised setting, the data may provide only group-level labels together with member bounding boxes, without reliable identity labels for each individual. This distinction is important because the availability of member identities directly influences how person-level and group-level representations can be learned.

The performance of group re-identification systems is commonly evaluated using ranking-based retrieval metrics. The most widely used metrics are the Cumulative Matching Characteristic (CMC) and mean Average Precision (mAP). The CMC curve measures the probability that at least one correct match appears within the top-k retrieved gallery results. In particular, Rank-1 accuracy is the proportion of queries for which the correct match is placed first in the ranking. It can be written as

$$\mathrm{CMC}(k) = \frac{1}{N_q} \sum_{i=1}^{N_q} \mathbf{1}(r_i \leq k), \tag{1.1}$$

where N_q is the number of query images, r_i is the rank position of the first correct match for the i-th query, and $\mathbf{1}(\cdot)$ is the indicator function.

Mean Average Precision provides a more comprehensive evaluation because it reflects both precision and recall over the full ranking list. For each query, the Average Precision (AP) is computed from the ranked retrieval results, and the final score is obtained by

averaging across all queries:

$$\mathrm{mAP} = \frac{1}{N_q} \sum_{i=1}^{N_q} \mathrm{AP}_i. \tag{1.2}$$

In the context of group re-identification, Rank-1 emphasizes whether the correct group match is retrieved first, whereas mAP evaluates the overall quality of the ranking when multiple relevant matches may exist in the gallery. Together, these metrics provide a standard quantitative basis for comparing group re-identification methods.

1.3 Research Object and Subject

The object of the research is video-based analysis of human groups in public spaces. The subject of the research is methods and models for representing, matching, and evaluating human groups across non-overlapping camera views.

1.4 Aim and Objectives

The aim of this diploma work is to develop a group re-identification system for analyzing human groups in public spaces.

To achieve this aim, the following objectives are defined:

1. to analyze existing approaches to person and group re-identification;
2. to identify the main technical challenges of group re-identification, including topology variation, occlusion, and member changes;
3. to design a system architecture for group detection, representation, matching, and evaluation;
4. to define an experimental methodology and performance metrics for assessing the proposed solution;
5. to summarize the practical limitations and directions for further development.

1.5 Contribution of the Work

The contribution of this diploma work consists in the structured formulation of group re-identification as a practical computer-vision task for public-space analysis and in the design of a system-oriented solution for this problem. In particular, the work contributes:

1. an academic analysis of why group re-identification cannot be reduced to conventional person re-identification;
2. a problem formulation that explicitly considers topology changes, member variation, and intra-group occlusion;
3. a clean system structure for group representation, cross-camera matching, and evaluation in surveillance scenarios;
4. a basis for further experimental comparison with existing group re-identification approaches.

1.6 Scientific Novelty and Practical Significance

The scientific relevance of the work lies in the system-level study of group re-identification as a task that extends beyond individual appearance matching toward collective representation and relational reasoning. The practical significance of the work is determined by the potential application of the developed approach in surveillance systems, smart-city platforms, and video analytics tools for public-space monitoring.

1.7 Structure of the Diploma Work

The diploma work consists of an introduction, five chapters, a conclusion, a bibliography, and appendices. Chapter 1 introduces the problem and research objectives. Chapter 2 reviews related work in person and group re-identification. Chapter 3 describes the proposed system design and methodology. Chapter 4 presents the experimental evaluation framework and results structure. Chapter 5 summarizes the main findings and conclusions.

Chapter 2

Literature Review

2.1 Person Re-Identification

Person re-identification provides the conceptual foundation for group re-identification. In the classical formulation, the task is to match images of the same pedestrian across non-overlapping cameras despite changes in illumination, pose, viewpoint, scale, and partial occlusion. Over time, the field has evolved from hand-crafted descriptors and metric-learning methods to deep convolutional and transformer-based architectures that learn discriminative appearance embeddings directly from data [2].

Despite this progress, person re-identification remains centered on the individual as the unit of analysis. This assumption becomes limiting when the practical target is a social group rather than a single pedestrian. A group is not simply a collection of independent person identities; it is a structured entity whose members interact, change relative positions, and may not remain fully visible across cameras. Therefore, group re-identification extends person-level matching by introducing collective representation, structural reasoning, and robustness to incomplete membership correspondence [2, 3].

2.2 Group Re-Identification

Research in group re-identification has evolved from relatively simple global descriptors to sophisticated deep models that attempt to capture the internal organization of human groups. Early methods typically represented a group image as a single visual entity and relied on hand-crafted descriptors designed to be robust to viewpoint or illumination

changes. Although these methods established the basic retrieval setting for group re-identification, they were limited by their inability to explicitly model member-level structure and dynamic group behavior [2].

Subsequent work introduced multi-grained and relation-aware strategies. Instead of relying exclusively on a single global descriptor, these methods attempted to describe groups at several levels, for example through individual members, subgroups, and the complete group image. This shift reflected an important insight: group identity is often preserved through a combination of local and global cues rather than through one holistic representation alone [7].

More recent approaches use graph neural networks, transformer architectures, and uncertainty-aware modeling to capture richer structural relationships. These methods have improved the ability of group re-identification systems to reason about context, topology, and ambiguity. However, a recurring limitation in many of these approaches is that they still treat group members too uniformly. In practice, not every member contributes equally to group identity, and not every spatial relation is equally stable across views [1].

2.3 Existing Methods and Research Trends

The main methodological directions in group re-identification can be summarized as follows.

2.3.1 Descriptor-Based and Global Matching Approaches

Early approaches focused on manually designed descriptors that summarized the overall visual characteristics of a group image. Examples include ratio-occurrence descriptors, covariance-style descriptors, salience-based patch selection, and sparse feature encoding schemes. These methods sought invariance to viewpoint and illumination changes, but they usually treated the group as a single holistic pattern. As a result, they were not sufficiently robust to strong topology changes, severe occlusion, or membership variation [2].

2.3.2 Multi-Grained Representation Learning

To overcome the limitations of purely global matching, multi-grained methods introduced representations at different structural levels. A representative example is the MGR framework, which models groups through fine-grained individuals, medium-scale subgroups, coarse substructures, and the full group image. The main benefit of such methods is improved flexibility: even if the global layout changes, some local or intermediate structures may still remain matchable. However, these methods still face difficulty when the reliability of different members varies significantly across views [7].

2.3.3 Graph-Based Context Modeling

Graph-based methods represent a group as a relational graph whose nodes correspond to members and whose edges encode contextual dependencies. MACG (Multi-Attention Context Graph) enriches person-level features through intra-group and inter-group attention mechanisms, while GCGNN (Group Context Graph Neural Network) constructs spatially informed neighborhood graphs and aggregates information according to member proximity and structural relations. These methods better capture interaction context than descriptor-based models, but their performance still depends on the quality and stability of the underlying member graph [3, 8].

2.3.4 Transformer and Topology-Aware Methods

Transformer-based approaches extend relation learning by modeling long-range dependencies and structural ambiguity more explicitly. For example, 3DT uses three-dimensional relational reasoning to reduce the ambiguity caused by 2D layout projections, while uncertainty-aware transformer methods attempt to represent groups as distributions rather than fixed deterministic samples. Such methods are more expressive than earlier approaches, but many still apply attention uniformly across members and may therefore fail to reflect the fact that some members are central to group identity while others are peripheral or unstable [4, 5, 9].

2.3.5 Weakly Supervised and Cross-Modality Methods

Another important trend is the study of weak supervision and cross-modality matching. Methods such as CPM-based matching emphasize permutation-invariant comparison and attempt to remain effective when precise person-level labels are not available.

Cross-modality variants extend the task to RGB-infrared settings for surveillance under day-night transitions. These directions improve practical applicability, but they also amplify the difficulty of robust member correspondence and group-level representation learning [6].

2.4 Comparative Analysis of Representative Methods

Table 2.1 summarizes representative approaches and their main limitations.

Paper	**Method**	**Key Idea**	**Weakness**
Early descriptor-based methods	Global descriptor matching	Represent the group image through hand-crafted global or patch-based descriptors	Weak robustness to topology changes, occlusion, and dynamic membership; limited structural reasoning
MGR	Multi-grained representation	Match groups at several structural levels, from individual members to the full group	Still does not explicitly distinguish stable and unstable members in a principled way
MACG	Context graph with attention	Model members as graph nodes and update features using contextual attention	Sensitive to graph quality; member importance is not strongly differentiated
GCGNN	Graph neural network	Use spatial neighborhood graphs and relation-aware aggregation for group modeling	Relational modeling is improved, but member contributions can still be treated too uniformly
3DT	Transformer-based topology modeling	Reduce 2D layout ambiguity through richer structural reasoning	More expressive than global methods, yet not fully aligned with intra-group role differentiation
Uncertainty-aware transformer methods	Distribution-based transformer learning	Model group variation and ambiguity through uncertainty-aware representations	Can capture variation, but may still distribute attention too broadly across all members
CPM-based methods	Permutation-invariant matching	Improve matching under layout disorder and weak supervision	Less effective when semantic role differences between members are critical
SIM	Social interaction modeling	Differentiate core and peripheral members and model socially meaningful intra-group roles	Depends on accurate estimation of member importance and interaction structure

Table 2.1: Representative methods in group re-identification and their main limitations

2.5 Research Gap

An important limitation of many existing methods is that they treat all members of a group as if they were equally informative and equally stable. This assumption appears

in different forms across the literature. Early global methods collapse the group into a single representation and therefore suppress member-level differences entirely. Even more advanced graph and transformer methods often learn relational structure without explicitly distinguishing between central, stable members and peripheral, unstable ones.

This limitation is significant because real human groups are not internally uniform. Social interaction patterns suggest that some members form the core of the group and remain relatively stable in position and participation, whereas others occupy more peripheral roles and are more likely to change position, become occluded, or disappear across camera views. When a model treats all members equally, it may allocate excessive attention to unstable or weakly informative members, which increases intra-class variation and reduces retrieval accuracy [1].

From this perspective, the development of SIM-style approaches marks an important conceptual step. Instead of assuming undifferentiated group structure, social interaction modeling explicitly accounts for intra-group differentiation. The key insight is that group identity should be learned not only from the presence of members, but also from their relative importance and functional role inside the social unit. This idea directly addresses a major weakness of earlier approaches and provides a strong foundation for the design of more robust group re-identification systems [1].

Thus, the literature indicates the following progression. Early methods based on global features provided a necessary starting point, but they were too coarse to model dynamic groups effectively. Transformer- and graph-based approaches improved structural reasoning, yet many of them still relied on implicitly uniform treatment of group members. SIM and related differentiation-oriented methods advance the field by recognizing that member roles inside a group are unequal and that this inequality must be reflected in the learned representation. This gap between undifferentiated modeling and socially informed group representation forms one of the central motivations of the present diploma work [1].

Chapter 3

Methodology

3.1 Overview of the Proposed Method

In this diploma work, the methodological basis of the group re-identification system is the Social Interaction Modeling (SIM) framework. The central idea of SIM is that a human group should not be treated as a uniform collection of members with equal importance. Instead, the group is modeled as a social interaction field in which different members play different roles. Some members form the stable core of the group, while others occupy more peripheral positions and are more likely to change location, become occluded, or disappear across camera views [1].

This assumption directly addresses one of the central limitations identified in the literature review. Many earlier methods either relied on global group features or used graph and transformer mechanisms that still allocated attention too uniformly across members. SIM attempts to solve this problem by introducing explicit intra-group differentiation and by using social priors to guide representation learning [1].

The overall SIM pipeline can be described as follows. First, a group image is represented through its detected members and their bounding boxes. Next, the social relations among members are estimated through the Social Interaction Calculation Module (SICM). On the basis of these interaction estimates, the Social Prior Attention Mechanism (SPAM) guides feature learning by assigning higher importance to socially central members. Finally, the Social Layout Variation Module (SLVM) models realistic topology changes by allowing different members to move with different degrees of freedom. In this way, the framework jointly addresses identity differentiation, member variation, and non-rigid

group layout changes [1].

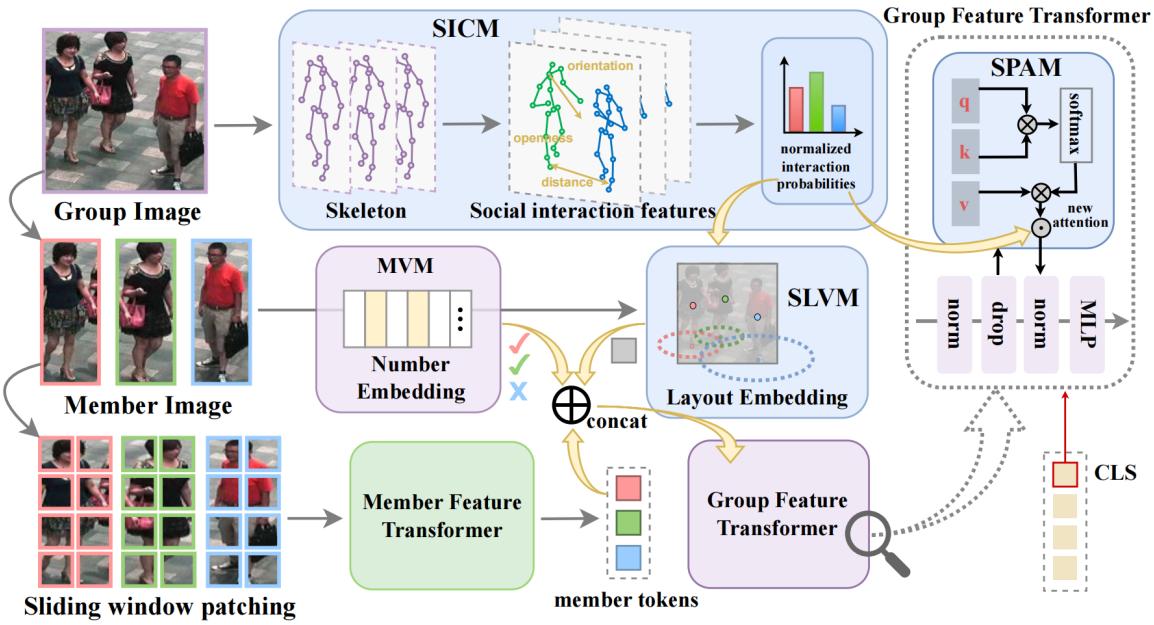

Figure 3.1: Pipeline of the SIM framework reproduced from the SIM paper and used as methodological context in this work [1].

3.2 Social Interaction Calculation Module

The Social Interaction Calculation Module (SICM) is responsible for quantifying the internal social relationships of a group. Its purpose is to convert visual observations of member arrangement into interaction probabilities that reflect how strongly each member belongs to the group structure. This module provides the basis for distinguishing core and peripheral members.

The interaction probability between two members i and j is derived from a social interaction intensity term. In simplified form, this relationship may be written as

$$p_{ij} = 1 - e^{-S_{ij}/\lambda}, \tag{3.1}$$

where S_{ij} denotes the interaction strength between members i and j, and λ controls the decay of interaction with respect to the underlying social field. In the more detailed SIM formulation, the interaction probability is generalized as

$$p_{ij} = 1 - \exp\left(-Sk_{ij}/\lambda\right)^b, \tag{3.2}$$

where Sk_{ij} is the intensity of the corresponding subfield and b is a shape parameter [1].

The key idea is that S_{ij} or Sk_{ij} is not an arbitrary score. It is constructed from visual primitives that describe socially meaningful relationships between group members. In particular, SIM relies on three components:

1. distance between members;
2. relative orientation;
3. pose-openness.

3.2.1 Distance Component

The distance term captures the spatial proximity between two members. In the SIM framework, it is computed from the bottom midpoints of the corresponding bounding boxes. Because distances measured directly in the image plane may be distorted by perspective effects, a perspective transformation is applied to improve geometric consistency across the scene. Intuitively, the smaller the distance between two members, the more likely they are to belong to the same social interaction field.

3.2.2 Orientation Component

The orientation term reflects whether two individuals are physically oriented toward one another in a manner consistent with social interaction. This quantity is derived from the angle between a member's orientation vector and the vector connecting that member to another pedestrian. If two members are facing in directions that support interaction, their interaction score should increase. If their orientations are inconsistent with mutual engagement, the interaction score should decrease.

3.2.3 Pose-Openness Component

The pose-openness term captures whether a member's posture is socially open or closed. In the SIM description, this variable is discretized into values such as open, neutral, or closed. Its role is to refine the interaction estimate by incorporating posture cues that indicate how available a person is for social engagement within the group [1].

3.2.4 Normalized Member Importance

After pairwise interaction probabilities are computed, SICM aggregates them to estimate the relative importance of each member inside the group. The average interaction probability of member i is written as

$$\bar{p}_i = \frac{1}{N-1} \sum_{j \neq i} p_{ij}, \tag{3.3}$$

where N is the number of members in the group. These values are then normalized:

$$\hat{p}_i = \frac{\bar{p}_i}{\sum_{i=1}^{N} \bar{p}_i}. \tag{3.4}$$

The normalized probability $\hat{p}_i$ expresses the relative social importance of each member. Members with high values are interpreted as core members, because they exhibit stronger and more stable interaction patterns with the rest of the group. Members with low values are interpreted as peripheral members and are more likely to correspond to weakly attached or unstable group elements [1].

From the perspective of topology handling, SICM is essential because it provides a principled way to identify which members are structurally central to the group. This makes it possible to treat group changes non-uniformly rather than assuming that every member contributes equally to the identity of the group.

3.3 Social Prior Attention Mechanism

The Social Prior Attention Mechanism (SPAM) uses the interaction probabilities produced by SICM to guide representation learning. Its purpose is to implement identity differentiation by assigning greater attention to core members and lower attention to peripheral members.

In a standard attention mechanism, the raw attention matrix is obtained from query-key interactions:

$$A_{\text{raw}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V, \tag{3.5}$$

where Q, K, and V denote the query, key, and value matrices, and d is the embedding dimension. In SIM, this attention process is modulated by the social priors estimated by

SICM. If the vector of normalized interaction probabilities is denoted by $\hat{P}_g$, then the socially weighted attention can be expressed as

$$A_{\text{social}} = A_{\text{raw}} \odot \hat{P}_g, \tag{3.6}$$

where $\odot$ denotes the Hadamard product [1].

The mathematical intuition is straightforward. If a member has a high social importance score, then its features are amplified during attention-based aggregation. If a member has a low importance score, its influence on the final group representation is suppressed. As a result, the learned representation is less sensitive to peripheral members that may disappear or become unreliable across camera views.

This mechanism is particularly important for handling member variation. Real groups are compositionally fluid, and peripheral members are often the first to leave the group or become inconsistent across observations. By concentrating learning on core members, SPAM reduces the effect of such noise and decreases the intra-class variation of group features. In this sense, SPAM acts as the identity-stabilizing component of the SIM framework [1].

3.4 Social Layout Variation Module

While SICM and SPAM focus on member importance and representation weighting, the Social Layout Variation Module (SLVM) addresses the problem of non-rigid group topology changes. Its role is to model the fact that different members do not move in the same way. In earlier augmentation strategies, a single affine transformation was often applied uniformly to the whole group. Such an approach is unrealistic because core members usually remain relatively stable, whereas peripheral members show larger positional shifts.

SLVM models this behavior by assigning a distinct position variation offset to each member:

$$\Delta d_i = \alpha \left(\sum_{j \neq i} \hat{p}_j (d_j - d_i) \right) + (1 - \alpha) r_i, \tag{3.7}$$

where α is a balancing coefficient, $d_j - d_i$ describes the spatial relationship between members, and r_i is a random perturbation term [1].

The final layout is then obtained by adding the displacement field to the original layout:

$$D_{\text{final}} = D_{\text{ori}} + \Delta D. \tag{3.8}$$

This formulation combines two sources of variation. The first is a structure-aware component, weighted by social importance, which encourages the layout evolution to remain consistent with the underlying group structure. The second is a random perturbation component that preserves diversity and prevents the model from overfitting to a narrow set of configurations.

The role of SLVM in handling topology changes is especially important. Core members, because they are strongly integrated into the social field, are constrained to smaller and more stable displacement ranges. Peripheral members, in contrast, are allowed greater spatial freedom. This produces more realistic topological variation during training and improves robustness to the non-rigid layout changes encountered in real surveillance scenarios [1].

3.5 Role of SIM in Handling Group Topology Changes

The effectiveness of SIM follows from the complementary interaction of its three modules. SICM identifies the social structure of the group through distance, orientation, and openness cues. SPAM translates this structure into attention-based feature weighting, thereby emphasizing members that are most representative of group identity. SLVM then uses the same social priors to model realistic spatial variation instead of applying uniform layout changes.

Together, these components reduce the intra-class distance of group representations by making the learned embedding more stable with respect to occlusion, partial member disappearance, and topology variation. In other words, SIM does not merely describe a group by its visible members; it models how those members participate in the social unit. This makes the framework particularly suitable for group re-identification in public spaces, where groups are dynamic, non-rigid, and observed under heterogeneous camera conditions [1].

Chapter 4

Experimental Evaluation

4.1 Evaluation Protocol

The experimental evaluation of group re-identification models must account for the fact that groups are non-rigid, compositionally dynamic, and subject to severe cross-camera appearance variation. For this reason, a meaningful experimental setup should specify the benchmark datasets, the retrieval metrics, the evaluation protocol, and the set of baseline methods used for comparison [2].

In the broader literature, group re-identification is commonly studied on datasets such as i-LIDS MCTS, Road Group, DukeMTMC Group, and CSG, as well as on more recent large-scale or cross-modality benchmarks. Small-scale datasets such as i-LIDS MCTS and DukeMTMC Group were important in the early development of the field because they established the feasibility of cross-camera group matching under real surveillance conditions. Road Group remains relevant as a challenging benchmark due to dense scenes, occlusion, and substantial layout variation. CSG is particularly important because it provides a larger and more diverse benchmark for studying modern group re-identification methods under more realistic retrieval settings [2, 3].

In this diploma work, the experimental study is conducted on two datasets: **CSG** and **RoadGroup**. These two benchmarks were selected because they complement one another. CSG provides a substantially larger evaluation environment, while RoadGroup captures difficult road-scene observations with strong occlusion and non-rigid group changes. As a result, their joint use allows the proposed model to be evaluated under both larger-scale and structurally challenging conditions [3, 6].

According to the recorded evaluation logs, the CSG test protocol used in this work includes 1,724 query images and 6,738 gallery images. The corresponding query split contains 699 group identities and 1,572 person identities, while the gallery split contains 5,713 group identities and 7,533 person identities. For RoadGroup, the evaluation protocol includes 81 query images and 81 gallery images, with 81 group identities in each split. These settings reflect the actual data configuration used during the experiments.

The evaluation pipeline follows the standard retrieval-based formulation of group re-identification. For each query group image, the model computes a feature representation and ranks all gallery images according to feature similarity. The final output is therefore a ranked list of candidate group matches for each query. In the present implementation, cosine similarity is used as the test metric, and the best checkpoint obtained during training is used for final evaluation.

4.2 Metrics

The primary evaluation metrics in this work are Rank-1 accuracy and mean Average Precision (mAP). These metrics are standard in person and group re-identification because they jointly assess ranking quality and retrieval completeness.

Rank-1 accuracy is derived from the Cumulative Matching Characteristic (CMC) curve and measures how often the correct match appears in the first position of the retrieval list. If r_i denotes the rank of the first correct match for query i, then the CMC at rank k is defined as

$$\mathrm{CMC}(k) = \frac{1}{N_q} \sum_{i=1}^{N_q} \mathbf{1}(r_i \leq k), \tag{4.1}$$

where N_q is the number of queries and $\mathbf{1}(\cdot)$ is the indicator function. In practice, Rank-1 is the most important special case because it directly reflects whether the system retrieves the correct group at the first attempt.

Mean Average Precision complements Rank-1 by evaluating the full ranking list. For each query, the Average Precision (AP) is computed from the ranked retrieval results, and the final mAP score is obtained as

$$\mathrm{mAP} = \frac{1}{N_q} \sum_{i=1}^{N_q} \mathrm{AP}_i. \tag{4.2}$$

While Rank-1 emphasizes the first correct retrieval, mAP captures the overall quality of ranking and is especially informative when multiple relevant matches may exist in the gallery.

In addition to these two principal metrics, the evaluation logs generated in the experiments also report Rank-5, Rank-10, and mINP. However, the main focus of the present work remains on Rank-1 and mAP because these are the most widely used summary indicators in the group re-identification literature [2].

4.3 Baseline Methods

To interpret the performance of the evaluated model, it is necessary to consider the main baseline families used in the literature. These baselines form a methodological progression from simpler to more expressive approaches.

The first group consists of traditional hand-crafted descriptor methods, which represent groups through global appearance or patch-based statistics. Such methods are historically important, but they are limited in their ability to handle severe topology changes and fluid group composition.

The second group includes appearance-based deep learning models such as MGR, MACG, and DotSCN. These methods introduce multi-grained or context-aware group representations and generally provide stronger discriminative power than descriptor-based approaches [3, 7, 10].

The third group includes graph-based and transformer-based methods such as GCGNN, uncertainty-aware transformer approaches, and SIM. These methods attempt to capture richer spatial relations, layout ambiguity, and member-level interactions. In the context of the present work, SIM is of particular importance because it introduces explicit differentiation between core and peripheral members, thereby addressing a key weakness of many earlier methods [1, 4, 5, 8].

In the current stage of this diploma work, the experimental results available in the repository correspond to the SIM-based model evaluated on CSG and RoadGroup. Therefore, the baseline discussion in this chapter serves primarily as methodological context for the interpretation of results, while direct quantitative comparison with alternative methods can be added later when reference scores are integrated into the

thesis.

4.4 Results and Analysis

The experiments show that the evaluated SIM-based model performs strongly on both datasets used in this work. On the CSG dataset, the model achieved Rank-1 = 91.71% and mAP = 91.58%. On the RoadGroup dataset, the model achieved Rank-1 = 90.12% and mAP = 92.74%. These values indicate that the model is capable of stable cross-camera group retrieval under both larger-scale and more structurally difficult conditions.

Table 4.1 summarizes the main quantitative results obtained in the present experiments.

Dataset	**Rank-1 (%)**	**Rank-5 (%)**	**Rank-10 (%)**	**mAP (%)**
CSG	91.71	96.58	97.45	91.58
RoadGroup	90.12	96.30	97.53	92.74

Table 4.1: Experimental results of the reproduced SIM-based model on CSG and RoadGroup

From an academic perspective, these results should be interpreted primarily as a successful reproduction and analysis of SIM behavior on the datasets available in this diploma project. The current repository contains verified experimental outputs for SIM on CSG and RoadGroup, including metric files, retrieval manifests, retrieval grids, and stored test features. However, it does not yet contain a complete set of locally reproduced baseline runs for alternative methods. For this reason, the present section focuses on the observed behavior of the SIM-based model itself and only makes qualitative comparisons to baseline families discussed in the literature review.

The recorded logs also report Rank-5 and Rank-10 values above 96% on both datasets, which suggests that the correct group match is usually present among the highest-ranked retrieval candidates even when it does not occupy the first position. From an application perspective, this is important because many surveillance and forensic use cases can still benefit from a short list of high-confidence candidates rather than requiring a strictly perfect top-1 prediction in every case.

The quantitative pattern of the results is meaningful. On CSG, the model achieves slightly higher Rank-1 accuracy, which indicates strong ability to place the correct match at the top of the ranking even in a larger and more diverse gallery environment. On RoadGroup, the model obtains slightly higher mAP, which suggests that the full ranked list remains highly coherent despite the challenging road-scene conditions, occlusions, and layout deformation typical of this dataset. In other words, the model remains reliable not only at the first retrieval position, but across the broader set of likely candidates.

Qualitative retrieval artifacts were also generated during evaluation. The result archives contain retrieval grids, per-query top-k visualizations, and retrieval manifests for both CSG and RoadGroup. These outputs provide visual evidence that the model is able to retrieve semantically consistent group matches despite non-rigid layout changes and appearance variation. The retrieval examples for CSG and RoadGroup are shown in Figure 4.1 and Figure 4.4, respectively. In the saved visualized subset, the selected queries exhibit perfect retrieval behavior, which is consistent with the strong aggregate metrics reported by the model.

Figure 4.1: Enlarged retrieval visualization for the CSG dataset using the reproduced SIM-based model, part 1.

Figure 4.2: Enlarged retrieval visualization for the CSG dataset using the reproduced SIM-based model, part 2.

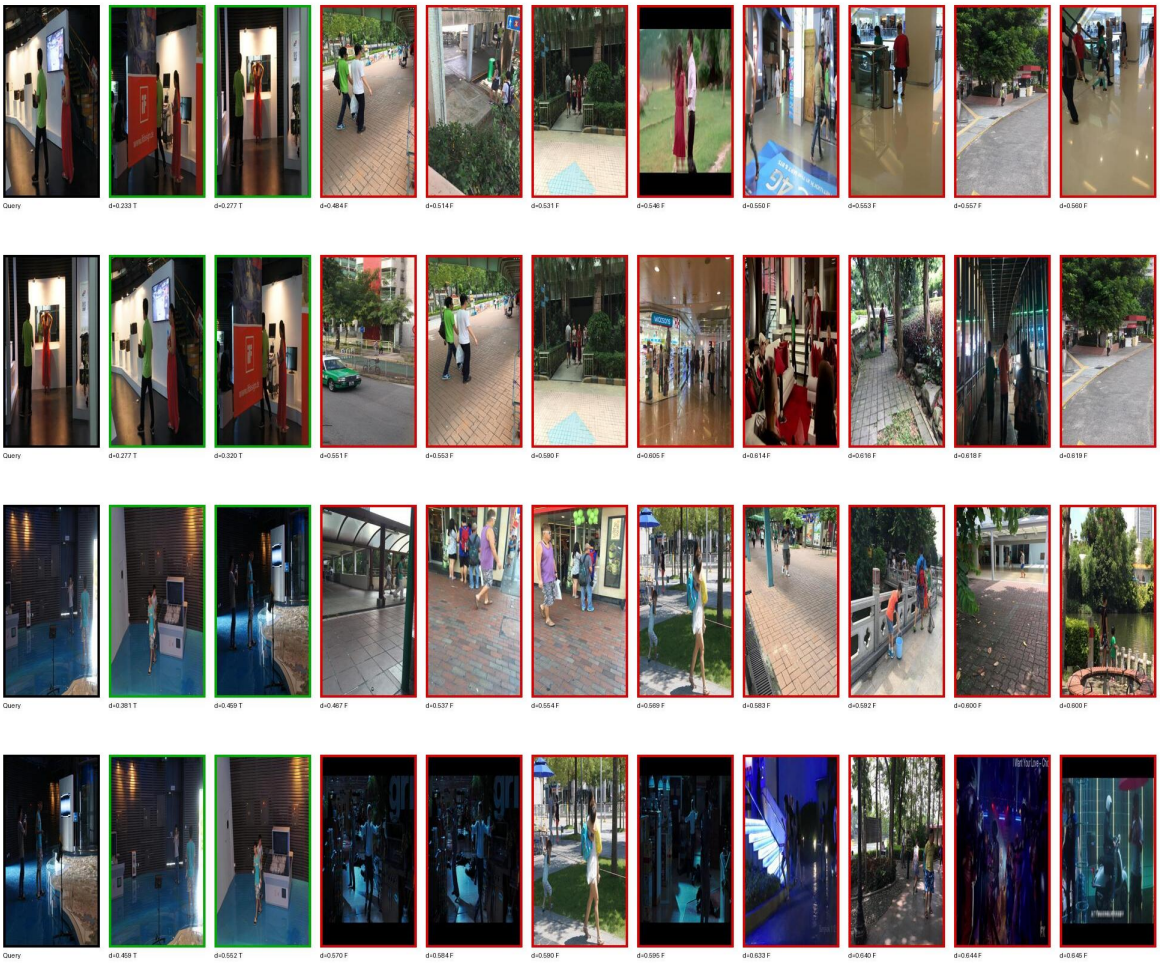

Figure 4.3: Enlarged retrieval visualization for the CSG dataset using the reproduced SIM-based model, part 3.

Figure 4.4: Enlarged retrieval visualization for the RoadGroup dataset using the reproduced SIM-based model, part 1. The image is rotated to improve readability in the document.

Figure 4.5: Enlarged retrieval visualization for the RoadGroup dataset using the reproduced SIM-based model, part 2. The image is rotated to improve readability in the document.

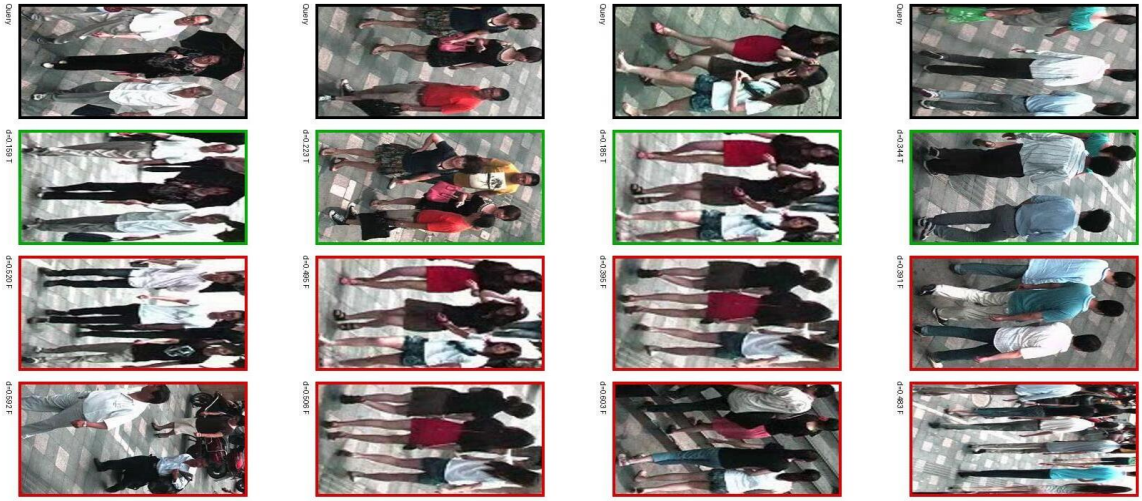

Figure 4.6: Enlarged retrieval visualization for the RoadGroup dataset using the reproduced SIM-based model, part 3. The image is rotated to improve readability in the document.

The behavior of the reproduced SIM model is also consistent with the methodological expectations established earlier in this thesis. SIM is designed to reduce the influence of unstable or weakly informative peripheral members by emphasizing socially central members and by modeling realistic non-rigid layout variation. High Rank-1 and mAP values on both datasets suggest that these mechanisms help preserve group identity under viewpoint change, partial occlusion, and member variation. This does not by itself prove numerical superiority over every baseline method, but it does support the conclusion that socially informed member differentiation produces a robust and practically effective group representation.

With respect to baseline comparison, the most defensible conclusion at the current stage is qualitative. Earlier global-feature methods are more likely to lose discriminative power when topology changes are substantial, while methods that treat all members uniformly may be more sensitive to disappearing or weakly related group members. SIM addresses these weaknesses directly through member differentiation. A strict numerical comparison against models such as MACG, GCGNN, 3DT, or UMSOT should be added only when the corresponding values are either reproduced locally or cited explicitly from the original papers with full source attribution.

The CSG and RoadGroup results should also be interpreted in relation to the distinct

nature of the datasets. CSG provides a larger and more diverse gallery environment, making strong performance on this dataset especially meaningful for scalability. RoadGroup, by contrast, is more compact but contains difficult road-scene images with substantial occlusion and deformation of group layout. The ability of the model to maintain high Rank-1 and mAP on both datasets supports the argument that SIM-style member differentiation is beneficial for handling realistic group variation.

4.5 Discussion

The main reason why SIM works is that it addresses a structural weakness of many earlier group re-identification methods: the assumption that all members of a group contribute equally to the final group identity. In real scenes, this assumption is rarely valid. Some members are central to the group and remain relatively stable across observations, whereas others are peripheral, transient, or only loosely associated with the group. By estimating social interaction strength and using that information to guide both attention and layout variation, SIM produces a group representation that is anchored to the most stable and informative members rather than being dominated by noisy or unstable ones.

More specifically, SIM works because its three modules solve complementary subproblems. SICM transforms geometric and pose-related cues into interaction probabilities, which serve as a compact estimate of member importance inside the group. SPAM then uses these probabilities to bias feature aggregation toward core members, thereby reducing the influence of peripheral members that may disappear between views. SLVM complements this process by modeling topological change in a non-uniform way, allowing peripheral members to vary more freely while preserving relative stability for the core. Together, these mechanisms reduce intra-class variation and make the learned representation more robust to non-rigid group behavior.

However, the effectiveness of SIM depends on several assumptions that may not always hold in real surveillance conditions. One important assumption is that social interaction can be reliably inferred from distance, orientation, and pose-openness cues. If pose estimation is inaccurate, if body orientation is ambiguous, or if bounding boxes are poorly localized, then the interaction probabilities computed by SICM may become unreliable. In such cases, the model may assign high importance to the wrong members or fail to detect the true core of the group.

Another difficulty arises in dense crowds. When the scene is highly congested, spatial proximity alone becomes a weaker indicator of true social relation. People who are physically close may simply be passing by one another rather than belonging to the same group. Under such conditions, the estimated interaction field can be distorted by incidental neighbors, which reduces the quality of both member differentiation and layout modeling. Thus, the social assumptions embedded in SIM are more reliable for coherent pedestrian groups than for highly entangled crowd scenes.

SIM may also become less reliable when group composition changes too abruptly. Its design assumes that at least some socially central members remain stable enough to preserve group identity across cameras. If these core members are fully occluded, leave the scene, or are replaced by newly appearing members, then the model loses the very anchors on which its representation depends. In such cases, even socially informed attention may not be sufficient to preserve cross-camera consistency.

There are also assumptions in the layout modeling that simplify real-world behavior. SLVM assumes that core members should generally move less than peripheral members, which is often reasonable but not universally true. For example, in fast-moving or goal-directed groups, leadership dynamics may cause central members to move aggressively while others reorganize around them. Similarly, road scenes or public events may induce abrupt collective movement patterns that do not follow the relatively smooth topological variation assumed by the model.

Finally, the very notion of a group is itself not always stable. In benchmark datasets, group identity is typically defined by annotation rules, often with a membership-overlap threshold. In real environments, however, the boundary between a true social group and a temporary spatial co-occurrence may be unclear. This means that the social interpretation used by SIM is meaningful, but still partially idealized. The model assumes that a visually inferred interaction field corresponds to a socially coherent unit, whereas real scenes may contain ambiguous or transitional formations.

For these reasons, SIM should be understood as a strong and well-motivated approach, but not as a universally reliable solution under all public-space conditions. Its main value lies in showing that socially informed intra-group differentiation improves group re-identification, while its main limitations highlight the need for more robust interaction estimation, better handling of dense crowds, and weaker dependence on accurate low-level cues such as pose and orientation.

4.6 Limitations

Several limitations should be noted when interpreting the current experimental results. First, the present evaluation is limited to two datasets, CSG and RoadGroup. Although these datasets are meaningful and complementary, a broader assessment on additional benchmarks such as i-LIDS MCTS or DukeMTMC Group would provide a more complete view of generalization.

Second, the currently available repository results do not yet include a unified quantitative comparison against multiple external baseline methods in the same experimental table. The literature review establishes the methodological baseline context, but a direct comparative table should be added later using reported benchmark values from prior work.

Third, group re-identification remains sensitive to strong occlusion, imperfect detection or annotation, and abrupt changes in group membership. Even a socially informed model such as SIM may still degrade when the core structure of the group is heavily disrupted or when the observed group contains substantial background interference.

Finally, the current experimental setup is tied to a specific implementation pipeline based on a patched SIM repository and Kaggle training notebooks. While this setup is sufficient for evaluation and demonstration, future work should further improve reproducibility by consolidating training, evaluation, and visualization into a cleaner and more standardized experimental framework.

Chapter 5

Conclusion

This diploma work considered the problem of group re-identification in public spaces. The relevance of this problem is determined by the growing practical importance of intelligent surveillance, smart-city analytics, and cross-camera understanding of collective pedestrian behavior. Unlike conventional person re-identification, group re-identification must operate on dynamic social units, which introduces additional complexity related to topology changes, intra-group occlusion, and variation in group membership.

The work showed that these challenges cannot be addressed adequately by simply extending single-person retrieval methods. A group is not a rigid visual object and not a uniform collection of equally informative members. For this reason, the thesis focused on approaches that model both individual appearance and the internal social structure of the group. The literature review demonstrated the evolution of the field from early global descriptor methods to graph-based and transformer-based architectures, and identified a key research gap: many existing methods still treat group members too uniformly.

To address this issue at the methodological level, the thesis adopted the Social Interaction Modeling (SIM) framework as the main model basis. SIM was analyzed through its three central components: the Social Interaction Calculation Module (SICM), the Social Prior Attention Mechanism (SPAM), and the Social Layout Variation Module (SLVM). Their joint role is to estimate member importance, emphasize socially central members during feature learning, and simulate realistic non-rigid layout changes. This makes the resulting group representation more robust to partial member disappearance, occlusion, and cross-camera structural variation.

The experimental part of the work was conducted on two datasets available in the current project: CSG and RoadGroup. The reproduced SIM-based pipeline achieved strong retrieval performance on both benchmarks. On CSG, the model reached Rank-1 = 91.71% and mAP = 91.58%. On RoadGroup, it achieved Rank-1 = 90.12% and mAP = 92.74%. These results show that the evaluated approach is capable of stable group retrieval under both larger-scale gallery conditions and more difficult road-scene layouts. In addition, retrieval visualizations confirmed that the model can produce semantically consistent matches for representative test queries.

An important outcome of this diploma work is not only the reporting of metric values, but also the analytical interpretation of model behavior. The study showed why SIM is effective: it reduces the influence of unstable peripheral members and anchors the group representation to more stable social structure. At the same time, the discussion highlighted important limitations. The method depends on the reliability of low-level cues such as pose, orientation, and bounding-box geometry; it may become less reliable in dense crowds or in cases where the social core of the group is heavily disrupted; and some of its assumptions about interaction and layout stability are simplified relative to real public-space behavior.

Thus, the main contribution of the diploma work is the structured study of group re-identification as a socially informed computer-vision problem, the adaptation of the SIM framework to the thesis context, and the experimental evaluation of this approach on CSG and RoadGroup. The work demonstrates that intra-group member differentiation is a meaningful direction for robust group representation and supports the broader conclusion that social context should play a central role in future group re-identification systems.

Future research can extend this work in several directions. First, the experimental study should be expanded to additional benchmarks such as i-LIDS MCTS and DukeMTMC Group to obtain a broader evaluation of generalization. Second, direct quantitative comparison with multiple baseline methods should be added in a unified experimental table. Third, the model itself can be improved through more reliable interaction estimation, stronger handling of dense crowds and ambiguous group boundaries, and tighter integration of structural and appearance-based cues. These directions would further strengthen the practical usefulness of group re-identification systems for real-world public-space analysis.

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Appendix A

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Appendix B

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